



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programme Division
M.Tech (Artificial Intelligence and Machine Learning)
Second Semester 2025-2026

Syllabus for Mid-Semester Examination (EC2) – ZG557

#	List of Topic Titles
1	- What is Artificial Intelligence: Acting Humanly, Thinking humanly, Thinking rationally, Acting Rationally - Foundations of AI, Brief Overview of Modern AI & Application Domains. - Intelligent Agents: Notion of Agents and Environments, Rationality - Nature of Environments, Structure of Agents
2	- Problem-Solving Agent - Problem Formulation - Examples - From Uninformed to Informed Search Algorithms - Notion of Heuristics - Informed Search Algorithms: Greedy best first search, A* search - Optimality of A*
3	- Heuristic Functions - Heuristic Accuracy & Algorithm performance - Admissible heuristics from relaxed problems, pattern databases & Experience - Use case: Learning Heuristics using Supervised Learning
4	- Local Search Algorithms & Optimisation Problems - Hill Climbing Search - Local Beam Search - Online Search Agents
5	- Genetic Algorithm - Ant Colony Optimization
6	- Neural Architecture Search - Use case: Neuro Evolution - Evolutionary computation to design or train neural networks, NEAT, DeepNEAT, CoDeepNEAT
7	- Adversarial Search - Searching to play games - Minimax Algorithm
8	- Static Evaluation Functions - Alpha-beta pruning - Monte Carlo Tree Search - Stochastic Games

The above topics from the Class Slides, textbooks, reference books, webinars, and research papers/online material are included in the exam.

***** ALL THE BEST *****